

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems.(L4,L5)

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SUBJECT:INTERNET PROGRAMMING

1. Define SOAP.

SOAP (Simple Object Access Protocol) is an XML-based messaging protocol used for exchanging structured information between applications over a network. It supports features such as security, reliability, and standardized error handling.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based language used to describe a web service. It specifies the available operations, messages, data types, communication protocol, and service endpoint.

3. What is an API?

API (Application Programming Interface) is a set of rules and methods that allows one software application to communicate with another application or service.

4. Define XML Schema.

XML Schema (XSD) is a language used to define the structure, elements, attributes, and data types allowed in an XML document. It is used to validate XML data.

5. What is a Web Service?

A **Web Service** is a software component that allows different applications to communicate and exchange data over a network using standard web technologies and protocols.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services using HTTP methods such as **GET, POST, PUT, and DELETE** to access and manipulate resources.

7. What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted and communicated between devices or applications over a network.

Examples: HTTP, HTTPS, FTP, TCP/IP.

8. Define Endpoint in Web Services.

An **endpoint** is the specific URL or network address through which a web service can be accessed.

Example:

`https://example.com/api/users`

9. What is a Service in Web Applications?

A **service** is a software component that provides specific functionality to other applications or clients through a defined interface.

Example: A payment service can process online payments.

10. What is the difference between SOAP and REST?

SOAP	REST
Protocol	Architectural style
Uses XML	Commonly uses JSON
More complex	Lightweight and simple
Supports WS-* standards	Commonly uses HTTP security mechanisms
Suitable for enterprise applications	Suitable for web/mobile applications

11. What is the role of WSDL in Web Services?

WSDL acts as a **contract** between a web-service provider and client. It tells the client **what operations are available, what data is required, how communication occurs, and where the service is located.**

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight data-interchange format. APIs commonly use JSON to send structured data between clients and servers.

Example:

```
{  
  "name": "Arun",  
  "age": 25  
}
```

13. What is HTTP and why is it used in Web Services?

HTTP (HyperText Transfer Protocol) is a communication protocol used to transfer data between clients and servers. Web services commonly use HTTP because it supports standard request methods such as **GET, POST, PUT, and DELETE.**

14. What is meant by Request and Response in Web Communication?

A **request** is a message sent by a client to a server asking for an operation or resource.

A **response** is the message returned by the server containing the requested data or the result of the operation.

Client → Request → Server

Client ← Response ← Server

15. What is the importance of XML in Web Services?

XML (Extensible Markup Language) provides a structured, platform-independent way to represent and exchange data. It is especially important in SOAP-based web services and is also used by WSDL and XML Schema.

16. What is the purpose of an API Endpoint?

An **API endpoint** identifies the specific URL through which a client accesses a particular API resource or operation.

Example:

GET /api/products/101

This endpoint can be used to retrieve product 101.

17. What is the difference between HTTP and HTTPS?

HTTP transfers data without TLS encryption, while **HTTPS (HTTP Secure)** uses TLS to encrypt communication between the client and server.

HTTP → Unencrypted communication

HTTPS → Encrypted + authenticated communication

HTTPS is preferred for sensitive information such as passwords, payments, and personal data.

18. What is meant by RESTful Services?

RESTful services are web services designed according to REST principles. They use resources identified by URLs and standard HTTP methods to perform operations.

Example:

GET /users → Retrieve users

POST /users → Create user

PUT /users/10 → Update user

DELETE /users/10 → Delete user

19. What is a Namespace in XML?

An **XML namespace** is used to uniquely identify XML elements and avoid naming conflicts when different XML vocabularies use the same element names.

Example:

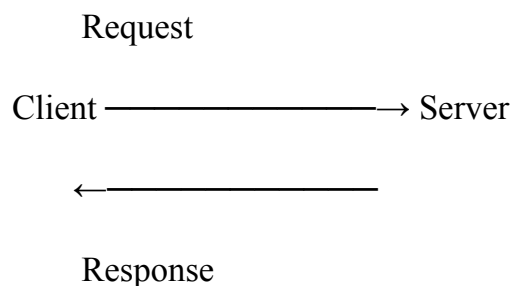
<soap:Envelope

xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

Here, soap is the namespace prefix.

20. What is the role of a Client and Server in Web Services?

The **client** initiates communication by sending a request, while the **server** receives the request, processes it, and sends a response.



Example: A mobile banking application is the client, while the banking web server processes its requests and returns account information.